Original Divide and Conquer Matrix Multiplication

$C11 = (A11*B11)+(A12*B21)$

$C12 = (A11*B12)+(A12*B22)$

$C21 = (A21*B11)+(A22*B21)$

$C22 = (A21*B12)+(A22B22)$

7 intermediary products

$P = (A11+A22)(B11+B22)$

$Q = (A21+A22)(B11)$

$R = (A11)(B12-B22)$

$S = (A22)(B21-B11)$

$T = (A11+A12)(B22)$

$U = (A21-A11)(B11+B12)$

$V = (A12-A22)(B21+B22)$

Reconstruction

$C11 = P+S-T+V$

$C12 = R+T$

$C21 = Q+S$

$C22 = P+R-Q+U$

Expansion

$C11 = [(A11+A22)(B11+B22)] + [(A22)(B21-B11)] - [(A11+A12)(B22)] + [(A12-A22)(B21+B22)]$

$C12 = [(A11)(B12-B22)] + [(A11+A12)(B22)]$

$C21 = [(A21+A22)(B11)] + [(A22)(B21-B11)]$

$C22 = [(A11+A22)(B11+B22)] + [(A11)(B12-B22)] - [(A21+A22)(B11)] + [(A21-A11)(B11+B12)]$

Algebraic Proof

C1:

$(A11B11)+(A12B21)=[(A11+A22)(B11+B22)] + [(A22)(B21-B11)] - [(A11+A12)(B22)] + [(A12-A22)(B21+B22)]$

Step 1: Foil the left side

$(A11*B11)+(A22*B11)+(A11*B22)+(A22*B22)+(A22*B21)-(A22*B11)-(A11*B22)-(A12*B22)+(A12*B21)-(A22*B21)+(A12*B22)-(A22*B22)$

Step 2: Find opposing terms and cancel

$(A11*B11)+(A22*B11)+(A11*B22)+(A22*B22)+(A22*B21)-(A22*B11)-(A11*B22)-(A12*B22)+(A12*B21)-(A22*B21)+(A12*B22)-(A22*B22)$

Step 3: Compare remaining terms to original

$(A11*B11) + (A12*B21) = (A11*B11) + (A12*B21)$ ✅

C2:
(A11B12)+(A12B22) = [(A11)(B12−B22)] + [(A11+A12)(B22)]
Step 1: Distribute
(A11*B12)-(A11*B22)+(A11*B22)+(A12*B22)
Step 2: Find opposing terms and cancel
(A11*B12)-(A11*B22)+(A11*B22)+(A12*B22)
Step 3: Compare remaining terms to original
(A11*B12)+(A12*B22)=(A11*B12)+(A12*B22)✅

C3:
(A21B11)+(A22B21) = [(A21+A22)(B11)] + [(A22)(B21−B11)]
Step 1: Distribute
(A21*B11)+(A22*B11)+(A22*B21)-(A22*B11)
Step 2: Find opposing terms and cancel
(A21*B11)+(A22*B11)+(A22*B21)-(A22*B11)
Step 3: Compare remaining terms to original
(A21B11)+(A22B21) = (A21B11)+(A22B21)✅

C4:
(A21B12)+(A22B22) = [(A11+A22)(B11+B22)] + [(A11)(B12−B22)] - [(A21+A22)(B11)] + [(A21−A11)(B11+B12)]
Step 1: Foil the left side
(A11*B11)+(A22*B11)+(A11*B22)+(A22*B22)+(A11*B12)-(A11*B22)-(A21*B11)-(A22*B11)+ (A21*B11)-(A11*B11)+(A21*B12)-(A11*B12)
Step 2: Find opposing terms and cancel
(A11*B11)+(A22*B11)+(A11*B22)+(A22*B22)+(A11*B12)-(A11*B22)-(A21*B11)-(A22*B11)+ (A21*B11)-(A11*B11)+(A21*B12)-(A11*B12)
Step 3: Compare to original
(A21*B12)+(A22*B22) = (A22*B22)+(A21*B12)✅